HARVARD UNIVERSITY

Pranav Rajpurkar, Ph.D.
Associate Professor of Biomedical Informatics, Harvard Medical School
10 Shattuck St 305, Boston, MA 02115
pranav_rajpurkar@hms.harvard.edu
https://rajpurkarlab.hms.harvard.edu
(650)-862-3446

Dear Editorial Team,

We are pleased to submit our manuscript titled "Grounding Radiology Reports in 3D Chest CT: Results of the ReXGrounding Challenge" for consideration in Medical Image Analysis. The manuscript reports the design, results and analysis of the ReXGrounding Challenge, held at MICCAI 2026, the first international challenge on segmenting the findings described in free-text radiology reports in 3D chest CT.

The Problem: Reports Describe Findings but Do Not Locate Them: Radiology reports describe the location of each finding in words, such as a nodule in the posterior segment of the left lower lobe, but they do not record its 3D extent, so referring physicians must search the scan for each finding again. Grounding each report sentence in the volume would let clinicians see, measure and track every reported finding, and would allow AI systems to generate reports whose findings can be verified against the image. Before the challenge, this task was largely unsolved. General text-promptable segmentation models such as SAT, SegVol and BiomedParse-V reached a Dice close to zero, and the best available model reached 0.285.

The ReXGrounding Challenge: The challenge asked teams to segment each finding of a report, given its free-text sentence and the CT volume. It was built on ReXGroundingCT and evaluated on 300 non-contrast chest CT scans with 582 findings, exhaustively annotated by radiologists. In total, 48 teams from 15 countries made 760 submissions during the challenge period, from 14 June to 14 September 2026. The paper goes beyond the leaderboard, with an analysis of ranking stability, performance by finding type, qualitative failure cases and a controlled experiment in which the best methods were run on deliberately edited sentences.

Key Findings: The challenge substantially advanced the state of the art. On the scans of the pre-challenge benchmark, the winning method reached a Dice of 0.390, against 0.285 before the challenge, and the best methods now match the agreement between two independent groups of radiologists in Dice and hit rate. The analysis also shows what remains unsolved. Models often find only some of the lesions of a multi-lesion finding, segment diffuse findings far less accurately than focal ones, and sometimes place the mask at a location other than the one the sentence names.

Relevance to Medical Image Analysis and Open Science: Medical Image Analysis has long been the home of challenge overview papers that define benchmarks for the community, and this manuscript provides the first such benchmark for grounding radiology language in 3D imaging. The data and evaluation code are public, and the leaderboard remains open with the same hidden test set, so that new methods can be compared directly with the entries reported here.

We believe this work will be of strong interest to the readership of Medical Image Analysis, from researchers developing vision-language and segmentation methods to those organizing future challenges. No form of this manuscript has been submitted or is under consideration elsewhere, and conflicts of interest are declared in the manuscript.

We look forward to your consideration.

Sincerely,

Pranav Rajpurkar, PhD
Associate Professor of Biomedical Informatics

Harvard Medical School